

SLURRY__FREE__ACID Modeling

Project Advancement Report

Data preparation, forecasting benchmarks, virtual sensing, and dashboard

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Current result: the target-anchored change model beats persistence at all four horizons. I also built a separate target-free virtual sensor for process-only data. The next required test is validation on another month.

Executive summary

I completed the following work:

- explored and checked one month of one-minute process data;
- prepared four forecast horizons and four feature-set hypotheses;
- benchmarked direct future-level prediction against persistence;
- improved forecasting by predicting the future target change;
- built a separate target-free virtual sensor;
- integrated the results into an interactive dashboard.

I kept three operational cases separate because their input requirements are different.

Case	Prediction contract	Executed result
Direct future level	Process measurements at or before t predict y_{t+h} .	Learned the operating level but did not beat persistence.
Target-anchored forecast	Process and target history predict $\Delta y_{t,h}$; reconstruct $\hat{y}_{t+h} = y_t + \widehat{\Delta y}_{t,h}$.	Ridge with the hybrid full set beat persistence by 19.8–34.9%.
Target-free virtual sensor	Process measurements only predict the current y_t .	Ridge–LightGBM blend achieved test RMSE 0.301 and $R^2 = 0.558$.

Current decision

- Use the target-anchored model when the current free-acid value is available.
- Use the virtual sensor for process-only data; it estimates the current value, not a future warning.
- Do not consider either model operationally validated until it is tested on another month.

1 Scope and data

Item	Executed scope
Dataset	44,640 one-minute observations covering 31 days in January 2026
Inputs	21 original process measurements
Target	SLURRY_FREE_ACID
Forecast horizons	1, 5, 10, and 15 minutes
Split policy	First 70% training, next 15% validation, final 15% test; no shuffle
Selection policy	Preprocessing, lag selection, filtering, hyperparameters, and blend weights selected without test fitting
Main benchmark	Persistence: $\hat{y}_{t+h} = y_t$ whenever the current target is available

Why chronological splitting: adjacent one-minute rows are dependent. Random splitting could place almost identical neighboring states in training and test and give overly optimistic results.

2 Work completed by notebook

Notebook	Purpose	Completed output
01	Data exploration	Timestamp, missingness, target behavior, sensor relationships, delayed behavior, and unusual-period assessment.
02	Feature preparation	Four horizons, process registry, selected lags, compact temporal and process-aware features, four feature sets, chronological splits, train-fitted preprocessing, and audit artifacts.
03	Direct benchmark	Mean and persistence baselines; Ridge, Extra Trees, and histogram gradient boosting across all 16 horizon/feature-set configurations.
04	Delta and transition study	Delta regression, target-history ablation, transition weighting, warning classification, regime errors, feature importance, and fitted research candidates.
05	Target-free virtual sensor	Process-only baselines, seven model families, validation-selected blend, drift diagnostics, empirical intervals, and saved inference bundle.
Dashboard	Interactive delivery	CSV upload, exploration, transition-window investigation, target-anchored forecasts, target-free estimates, drift flags, and label-dependent evaluation.

3 Feature-preparation methodology

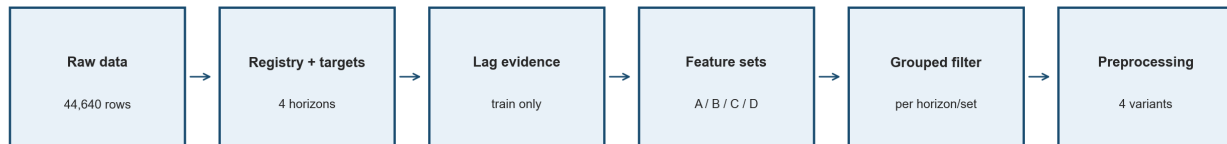


Figure 1: Executed feature-preparation pipeline. Every statistical decision that can leak target information is restricted to training data.

Targets and leakage controls

I created the direct targets:

$$y_{t+h} = \text{SLURRY_FREE_ACID}_{t+h}, \quad h \in \{1, 5, 10, 15\}.$$

- Every input at timestamp t uses data from t or earlier.
- I removed the first 30 rows because their required history does not exist.
- Later sensor gaps receive missingness indicators and training-median imputation.

Training-only lag selection

I tested lags of 1, 5, 10, 15, 30, and 60 minutes and retained at most three per sensor using:

- absolute lagged correlation with the one-minute future target;
- mutual information;
- sign and magnitude stability across three chronological training blocks;

- a 0.15 score bonus for a process-response prior.

I then added two rolling means, one rolling standard deviation, a five-minute difference, and a ten-minute slope per source variable.

Process-aware engineering

I added:

- total acid flow and acid-line imbalance;
- acid-to-phosphate ratio;
- confirmed total slurry volumetric flow;
- slurry-density proxy;
- thermal gradients and a steam-condition proxy;
- drying-intensity and recycle-to-production proxies;
- flow coefficients of variation.

Validation boundary

- Total slurry volumetric flow is confirmed as the sum of the two m^3/h signals.
- Total acid flow is confirmed additive, but its unit remains unresolved.
- Density, energy, drying, and recycle features remain proxies, not verified physical quantities.

Numerically risky process features use $\epsilon = 10^{-6}$, replace infinities with missing values, and are clipped using the training-period 0.1th and 99.9th percentiles.

Feature-set hypotheses and final dimensions

Set	Hypothesis	Candidates	Retained
A – Raw	Current measurements may be sufficient.	21	21
B – Temporal	Selected histories and recent dynamics add information.	210	162
C – Process-aware	Interpretable balances, ratios, gradients, and stability measures add information.	58	58
D – Full	Temporal and process-relationship evidence are complementary.	226	178

Correlation filtering

- Applied inside each original-sensor family at $|r| > 0.98$ using training data only.
- Priority: raw value, selected lag, rolling mean, difference, slope, then variability.
- Process-aware features and missingness indicators are protected from blanket removal.

4 Model training and selection

Controlled model families

Model	Role	Main executed choices
Ridge	Regularized linear reference for correlated features	Standard scaling; $\alpha \in \{1, 10\}$
PLS	Industrial latent-variable soft-sensor reference	5 or 10 components; pre-standardized inputs
KNN	Local historical-state hypothesis	25 or 75 distance-weighted neighbors
Extra Trees	Nonlinear thresholds and interactions	30–60 trees; leaf-size and feature-fraction alternatives
HistGradient	Efficient nonlinear boosting	50–80 iterations; controlled learning rates and leaf counts
XGBoost	Regularized boosted-tree reference	150 trees; depths 4 or 6; row and feature subsampling 0.8
LightGBM	Leaf-wise boosted-tree reference	150 trees; 15 or 31 leaves; minimum child size 40
Weighted blend	Complementary linear and nonlinear errors	Validation grid in 0.1 weight increments

Selection procedure

1. Fit each candidate on training data.
2. Rank candidates using validation RMSE.
3. Keep the lowest-validation-RMSE candidate.
4. Evaluate it once on the untouched test period.

I did not select models using test performance. The reported test models were not refitted on training plus validation.

5 Direct future-level benchmark

The first experiment learned

$$X_{\leq t} \longrightarrow \hat{y}_{t+h}.$$

Validation selected Feature Set D with Ridge at every horizon.

Horizon	MAE	RMSE	R^2	vs. persistence
1 min	0.269	0.332	0.465	-67.9%
5 min	0.275	0.338	0.445	-35.1%
10 min	0.280	0.344	0.426	-22.6%
15 min	0.304	0.378	0.306	-27.5%

Table 1: Untouched test performance for the validation-selected direct-level model. Negative improvement means worse than persistence.

Conclusion

- The model learned the broad operating level.
- It remained worse than carrying the current target forward.
- I therefore changed the task from predicting the level to predicting its future movement.

6 Target-anchored change forecasting

I reframed the task as:

$$\Delta y_{t,h} = y_{t+h} - y_t, \quad \hat{y}_{t+h} = y_t + \widehat{\Delta y}_{t,h}.$$

I compared four input families:

- 21 raw process measurements;
- 178 full exogenous features;
- 10 current/past target-history features;
- a 188-feature hybrid of full process information and target history.

Ridge and Extra Trees received equal two-candidate tuning budgets. Validation selected the 188-feature hybrid Ridge with $\alpha = 10$ at every horizon.

Horizon	MAE	RMSE	R^2	vs. persistence
1 min	0.102	0.151	0.889	+23.2%
5 min	0.113	0.169	0.862	+32.3%
10 min	0.126	0.182	0.839	+34.9%
15 min	0.176	0.237	0.726	+19.8%

Table 2: Untouched test performance after future-level reconstruction.

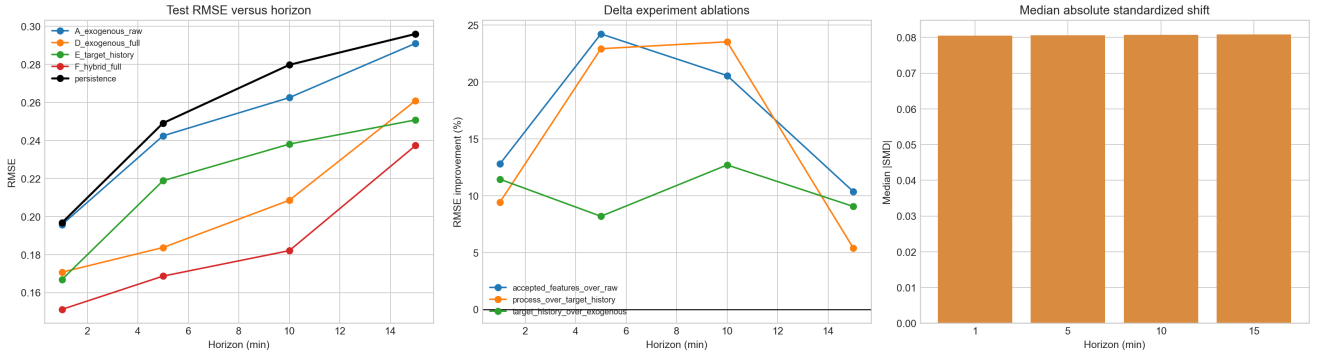


Figure 2: Delta performance, ablation gains, and standardized distribution shift. The hybrid full experiment gives the lowest RMSE across horizons.

Ablation evidence.

- Accepted engineered features improved over raw process inputs by 10.3–24.2%, depending on horizon.
- Target history improved over exogenous full inputs by 8.2–12.7%.
- Process/full information improved over target history alone by 5.4–23.5%.
- One minute gives the lowest absolute error.
- Ten minutes gives the largest relative improvement over persistence.
- The final horizon must be chosen using both error and required reaction time.

Transition and warning experiments

- Candidate transitions use the training 90th percentile of $|\Delta y_{t,h}|$.
- Extreme candidates use the 95th percentile.
- These are statistical labels, not confirmed plant events.
- Giving transition rows four times the training weight slightly improved some transition errors but worsened overall RMSE, so I did not retain that model.

Horizon	Precision	Recall	F1	False-warning rate
1 min	0.141	0.478	0.218	0.288
5 min	0.155	0.491	0.235	0.264
10 min	0.161	0.512	0.245	0.267
15 min	0.142	0.404	0.210	0.259

Decision: I kept the logistic transition classifier as a diagnostic only. Its precision is too low and its false-warning rate is too high for an operational alarm.

7 Target-free virtual sensor

The third case estimates the current target without target input:

$$X_{\leq t} \longrightarrow \hat{y}_t.$$

- Inputs: the 21 original process measurements and accepted Notebook 02 derivatives.
- Forbidden inputs: current target, target history, and future-target columns.
- Comparison: 14 experiments covering seven model families and two feature sets.

Validation selected the two-member blend:

$$\hat{y}_t = 0.5\hat{y}_t^{\text{Ridge,D}} + 0.5\hat{y}_t^{\text{LightGBM,D}}.$$

Test metric	Training-mean baseline	Selected blend
MAE	0.355	0.249
RMSE	0.454	0.301
R^2	approximately 0	0.558
Median absolute error	—	0.229
90th-percentile absolute error	—	0.459
95th-percentile absolute error	—	0.536

Table 3: Target-free current-time soft-sensor performance. RMSE improves by approximately 33.6% over the training-mean baseline.

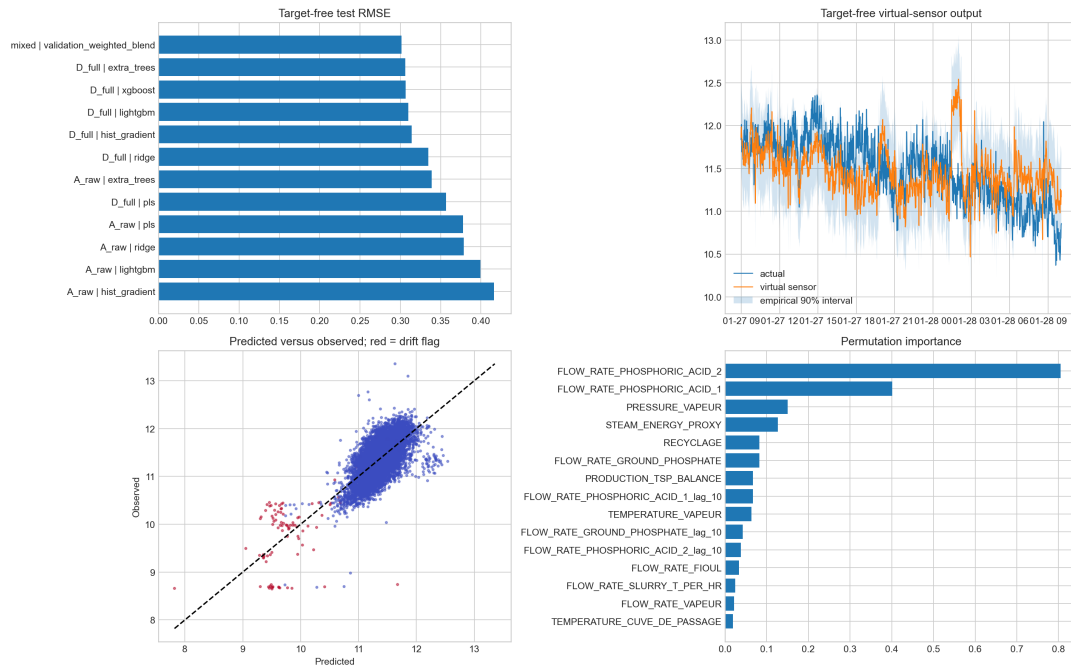


Figure 3: Target-free model comparison, example time series, predicted-versus-observed behavior, and validation permutation importance. Importance indicates association, not causality.

- The empirical 90% interval achieved 89.7% test coverage with mean width 1.015.
- The drift score flags rows beyond the training 99th percentile.
- Only 1.37% of test rows were flagged, but errors were substantially higher on those rows.

Uncertainty limitation: the interval width was calibrated from individual-Ridge validation residuals, while the final point estimate is a Ridge-LightGBM blend. I therefore treat it as an empirical diagnostic, not a formal coverage guarantee.

8 Interactive dashboard

I built the dashboard to make the notebook outputs usable without rerunning notebook cells. It supports:

- default-month analysis or CSV upload;
- signal trajectories, missingness, pairwise relationship exploration, and dataset observations;
- candidate multivariate process-change windows and their main sensor drivers;
- later target response and prediction-error comparison when labels are available;
- target-anchored forecasts at 1, 5, 10, and 15 minutes;
- target-free virtual-sensor estimates, uncertainty bands, and drift flags;
- report generation and cached preprocessing/model loading.

Important: the green multivariate process-change score measures joint standardized sensor movement. It does not use future target change and does not confirm a startup, shutdown, fault, or abnormal target event.

9 What is established and what is not

Supported by executed evidence	Not yet established
Chronological, leakage-controlled preparation across four horizons and four feature sets	Generalization to another month, season, product grade, or long-term sensor condition
Target-anchored delta forecasting beats persistence at every tested horizon	Operational warning performance during confirmed plant events
Temporal, process-aware, and target-history evidence provide measurable ablation gains	Causal influence or safe operator interventions
Target-free current-time estimation materially beats constant baselines	Target-free 5–15 minute early-warning performance
Input drift is associated with weaker prediction support	Guaranteed uncertainty coverage outside the observed training distribution

10 Process-validation requirements

I still need process-engineer confirmation for:

1. Confirm units of phosphoric-acid flow, ground-phosphate flow, fuel-oil flow, steam flow, wash-liquid flow, and recycle.
2. Define the exact material or process stream represented by RECYCLAGE.
3. Confirm whether FLOW_RATE_SLURRY_T_PER_HR represents the same combined stream as total slurry volumetric flow.
4. Identify confirmed startups, shutdowns, cleaning, maintenance, grade changes, and process disturbances.
5. Confirm which variables are genuinely controllable before any what-if or intervention recommendation.
6. Define acceptable prediction error, warning lead time, false-warning cost, and missed-event cost.

11 Recommended next work

1. **Freeze the current pipelines and test a later labeled month.** Report performance without retuning first; this is the highest-priority generalization test.
2. **Validate regimes against plant records.** Replace statistical “candidate” labels with confirmed event categories where possible.
3. **Recalibrate blend uncertainty.** Use out-of-sample blend residuals rather than individual-Ridge residuals.
4. **Run rolling-origin validation across multiple months.** Repeatedly train on the past and predict the next chronological block.
5. **Define the operational target.** Choose among current-time virtual sensing, target-anchored forecasting, and target-free future warning based on actual sensor availability.
6. **Only then refine models.** Increase tuning budgets or test sequence models only if multi-month evidence shows a remaining, practically important gap.

Current conclusion

What I have completed: data exploration, leakage-controlled feature preparation, direct and delta forecasting benchmarks, transition diagnostics, a target-free virtual sensor, and an interactive dashboard. The hybrid Ridge is the strongest forecast when the current target is available. The Ridge–LightGBM blend is the strongest target-free current estimator so far.